HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

Geometry and Trigonometry

06

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 C

C. $x - 4^2 + y - 9^2 = 16$

Choice C is correct. The graph of the equation $x - h^2 + y - k^2 = r^2$ in the xy-plane is a circle with center h, k and a radius of length r. The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the xy-plane intersects the y-axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the y-axis must be equal to the length of the circle's radius. It follows that the x-coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $x - h^2 + y - k^2 = r^2$ is a circle that intersects the y-axis at exactly one point, then $r = h$ must be true. The equation in choice C is $x - 4^2 + y - 9^2 = 16$, or $x - 4^2 + y - 9^2 = 4^2$. This equation is in the form $x - h^2 + y - k^2 = r^2$, where $h = 4$, $k = 9$, and $r = 4$, and represents a circle in the xy-plane with center $4, 9$ and radius of length 4. Substituting 4 for r and 4 for h in the equation $r = h$ yields $4 = 4$, or $4 = 4$, which is true. Therefore, the equation in choice C represents a circle in the xy-plane that intersects the y-axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the y-axis at any point.

Choice B is incorrect. This is an equation of a circle that intersects the x-axis, not the y-axis, at exactly one point.

Choice D is incorrect. This is the equation of a circle with the center located on the y-axis and thus intersects the y-axis at exactly two points, not exactly one point.

Q2 C

C. $\cos66°^2 + \cos24°^2$

Choice C is correct. The sine of an angle is equal to the cosine of its complementary angle. Since angles with measures 24° and $66°$ are complementary to each other, $\sin24°$ is equal to $\cos66°$ and $\sin66°$ is equal to $\cos24°$. Substituting $\cos66°$ for $\sin24°$ and $\cos24°$ for $\sin66°$ in the given expression yields $\cos66°\cos66° + \cos24°\cos24°$, or $\cos66°^2 + \cos24°^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3 156

The correct answer is 156. In the figure shown, the sum of the measures of angle UVS and angle RVS is 180°. It's given that the measure of angle RVS is 41°. Therefore, the measure of angle UVS is 180 - 41°, or 139°. The sum of the measures of the interior angles of a triangle is 180°. In triangle UVS, the measure of angle UVS is 139° and it's given that the measure of angle VST is 29°. Thus, the measure of angle VUS is 180 - 139 - 29°, or 12°. It's given that $RT = TU$. Therefore, triangle TUR is an isosceles triangle and the measure of VUS is equal to the measure of angle TRU. In triangle TUR, the measure of angle VUS is 12° and the measure of angle TRU is 12°. Thus, the measure of angle UTR is 180 - 12 - 12°, or 156°. The figure shows that the measure of angle UTR is $x°$, so the value of x is 156.

Q4

135/8

Accept also: `16.87`, `16.88`

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2, or 16, times the area of triangle DEF. It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF. It follows that 270 is 16 times a, or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

Q5

B

B. $(x - 16)^2 + (y - 17)^2 = 49k^2$

Choice B is correct. The equation of a circle in the xy-plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r. It's given that this circle has a center at $(16, 17)$ and a radius of $7k$. Substituting 16 for h, 17 for k, and $7k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 16)^2 + (y - 17)^2 = (7k)^2$, or $(x - 16)^2 + (y - 17)^2 = 49k^2$. Therefore, the equation that represents this circle is $(x - 16)^2 + (y - 17)^2 = 49k^2$.

Choice A is incorrect. This equation represents a circle with radius $7\sqrt{k}$, not $7k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{7}\,k$, not $7k$.

Q6

D

D. $\frac{18}{\sin Q}$

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS, an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of QS.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7

83

The correct answer is 83. It's given that in the figure, $AC = CD$. Thus, triangle ACD is an isosceles triangle and the measure of angle CDA is equal to the measure of angle CAD. The sum of the measures of the interior angles of a triangle is $180°$. Thus, the sum of the measures of the interior angles of triangle ACD is $180°$. It's given that the measure of angle ACD is $104°$. It follows that the sum of the measures of angles CDA and CAD is $180 - 104°$, or $76°$. Since the measure of angle CDA is equal to the measure of angle CAD, the measure of angle CDA is half of $76°$, or $38°$. The sum of the measures of the interior angles of triangle BDE is $180°$. It's given that the measure of angle EBC is $45°$. Since the measure of angle BDE, which is the same angle as angle CDA, is $38°$, it follows that the measure of angle DEB is $180 - 45 - 38°$, or $97°$. Since angle DEB and angle AEB form a straight line, the sum of the measures of these angles is $180°$. It's given in the figure that the measure of angle AEB is $x°$. It follows that $97 + x = 180$. Subtracting 97 from both sides of this equation yields $x = 83$.

Q8

12

The correct answer is 12. The volume, V, of a right square prism can be calculated using the formula $V = s^2h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is 2,016 cubic units and the height is 14 units. Substituting 2,016 for V and 14 for h in the formula $V = s^2h$ yields $2{,}016 = s^2 14$. Dividing both sides of this equation by 14 yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is 12.

Q9

C

C. $10, 2$

Choice C is correct. It's given that the circle has its center at $-1, 1$ and that line t is tangent to this circle at the point $5, -4$. Therefore, the points $-1, 1$ and $5, -4$ are the endpoints of the radius of the circle at the point of tangency. The slope of a line or line segment that contains the points a, b and c, d can be calculated as $\frac{d-b}{c-a}$. Substituting $-1, 1$ for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-1}{5--1}$, or $-\frac{5}{6}$. Thus, the slope of this radius is $-\frac{5}{6}$. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It follows that line t is perpendicular to the radius at the point $5, -4$, so the slope of line t is the negative reciprocal of the slope of this radius. The negative reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line t is $\frac{6}{5}$. Since the slope of line t is the same between any two points on line t, a point lies on line t if the slope of the line segment connecting the point and $5, -4$ is $\frac{6}{5}$. Substituting choice C, $10, 2$, for a, b and $5, -4$ for c, d in the expression $\frac{d-b}{c-a}$ yields $\frac{-4-2}{5-10}$, or $\frac{6}{5}$. Therefore, the point $10, 2$ lies on line t.

Choice A is incorrect. The slope of the line segment connecting $0, \frac{6}{5}$ and $5, -4$ is $\frac{-4-\frac{6}{5}}{5-0}$, or $-\frac{26}{25}$, not $\frac{6}{5}$.

Choice B is incorrect. The slope of the line segment connecting $4, 7$ and $5, -4$ is $\frac{-4-7}{5-4}$, or -11, not $\frac{6}{5}$.

Choice D is incorrect. The slope of the line segment connecting $11, 1$ and $5, -4$ is $\frac{-4-1}{5-11}$, or $\frac{5}{6}$, not $\frac{6}{5}$.

Q10 66

The correct answer is 66. It's given that each vertex of the rectangle lies on the circumference of the circle. Therefore, the length of the diameter of the circle is equal to the length of the diagonal of the rectangle. The diagonal of a rectangle forms a right triangle with the shortest and longest sides of the rectangle, where the shortest side and the longest side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. Let s represent the length, in units, of the shortest side of the rectangle. Since it's given that the diagonal is twice the length of the shortest side, $2s$ represents the length, in units, of the diagonal of the rectangle. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $a^2 + b^2 = c^2$. Substituting s for a and $2s$ for c in this equation yields $s^2 + b^2 = 2s^2$, or $s^2 + b^2 = 4s^2$. Subtracting s^2 from both sides of this equation yields $b^2 = 3s^2$. Taking the positive square root of both sides of this equation yields $b = s\sqrt{3}$. Therefore, the length, in units, of the rectangle's longest side is $s\sqrt{3}$. The area of a rectangle is the product of the length of the shortest side and the length of the longest side. The lengths, in units, of the shortest and longest sides of the rectangle are represented by s and $s\sqrt{3}$, and it's given that the area of the rectangle is $1{,}089\sqrt{3}$ square units. It follows that $1{,}089\sqrt{3} = ss\sqrt{3}$, or $1{,}089\sqrt{3} = s^2\sqrt{3}$. Dividing both sides of this equation by $\sqrt{3}$ yields $1{,}089 = s^2$. Taking the positive square root of both sides of this equation yields $33 = s$. Since the length, in units, of the diagonal is represented by $2s$, it follows that the length, in units, of the diagonal is 233, or 66. Since the length of the diameter of the circle is equal to the length of the diagonal of the rectangle, the length, in units, of the diameter of the circle is 66.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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